

*MA.8.GR.1.3***Benchmark**

MA.8.GR.1.3 Use the Triangle Inequality Theorem to determine if a triangle can be formed from a given set of sides. Use the converse of the Pythagorean Theorem to determine if a right triangle can be formed from a given set of sides.

Connecting Benchmarks/Horizontal Alignment

- MA.8.NSO.1.7
- MA.8.AR.2.2

Terms from the K-12 Glossary

- Converse of the Pythagorean Theorem

Vertical Alignment**Previous Benchmarks**

- MA.5.GR.1.1

Next Benchmarks

- MA. 912.GR.1.3

Purpose and Instructional Strategies

In grade 5, students classified triangles based on their angle measures and their side lengths. In grade 8, students use the Triangle Inequality Theorem and Pythagorean Theorem to determine whether triangles, or right triangles, can be formed from a given set of sides. In Geometry, students will extend this understanding to prove relationships and theorems about triangles.

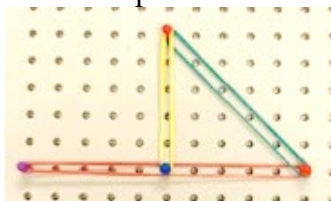
- Instruction includes modeling and drawing triangles with different side lengths to determine if they can make a triangle to help in conceptual understanding. Students can physically construct triangles with manipulatives such as straws, sticks, string or geometry apps prior to using rulers (*MTR.2.1*).
- Exploration should involve giving students three side measures to determine if a triangle can be made. Through discussion of their exploration results, students should conclude that triangles cannot be formed by any three arbitrary side measures.
 - For example, if students are given 4, 5 and 10, they should conclude that it does not form a triangle.
 - Through charting, students should realize that for a triangle to result, the sum of any two side lengths must be greater than the third side length. This can be charted in a table like the one below.

Length 1 (smallest)	Length 2 (middle)	Length 3 (largest)	Triangle?	$L1 + L2$

- Once students understand the Triangle Inequality Theorem, they can apply their knowledge to the converse of the Pythagorean Theorem. In work with the previous benchmark, students verify using a model that the sum of the squares of the legs is equal to the square of the hypotenuse in a right triangle. Students should also understand that if the sum of the squares of the 2 smaller legs of a triangle is equal to the square of the third leg, then the triangle is a right triangle.

Common Misconceptions or Errors

- Students may incorrectly think that the Triangle Inequality Theorem only applies to right triangles due to the work with the Pythagorean Theorem. Discussion of the two theorems and examples will help with this misconception.
- Students may incorrectly believe endpoints of the sides of the triangle do not have to meet at a vertex.
 - For example, students will attempt to make a triangle such as the example below.



Strategies to Support Tiered Instruction

- Instruction includes co-creating a graphic organizer to highlight key differences and use of the Pythagorean Theorem and the Triangle Inequality Theorem.
- Instruction includes the use of manipulatives, such as angle legs and geometric software, to allow for students to explore the similarities and differences between the Pythagorean Theorem and the Triangle Inequality Theorem.
- Teacher provides instruction on the definition of a triangle and allows for students to explore various side lengths using geometric software or manipulatives to determine if the lengths form a triangle.

Instructional Tasks

Instructional Task 1 (MTR.6.1, MTR.7.1)

The following side lengths, in meters, were given to a carpenter to build a front porch with a triangular design. The carpenter needs to determine which set of lengths will make a triangle to be able to use it in his design.

- Option 1: Side lengths: 4, 4, 8
- Option 2: Side lengths: 6, 8, 10
- Option 3: Side lengths: 6, 6, 13
- Option 4: Side lengths: 5, 6, 10

Part A. Which of the options would create a triangle for his design?

Part B. The homeowner would like the porch to be in the shape of a right triangle. Will the carpenter be able to use any of the given options?

Part C. For any option that does not form a triangle, what side length could be changed to form a triangle? Explain your answer.

Instructional Items

Instructional Item 1

Can the side lengths of a triangle be 2, 4 and 8? Justify your answer.

Instructional Item 2

John drew a triangle with side lengths of 5, 12 and 13. His friend, Bryan, looked at it and asked John if it is a right triangle. John's response was yes. Explain or show how John can prove to Bryan that the triangle is a right triangle.

**The strategies, tasks and items included in the B1G-M are examples and should not be considered comprehensive.*